A Comprehensive Guide to Performance Metrics in Machine Learning

Sitemap

Performance metrics play a crucial role in evaluating the effectiveness and accuracy of machine learning models. They provide insights into a model's predictive capabilities and help measure its performance across various tasks. In this blog post, we'll explore the significance of performance metrics, when to use them, and delve into popular metrics such as precision, recall, accuracy, and F-beta score, along with their formulas and examples.

Understanding Performance Metrics:

Importance:

- Performance metrics quantify the quality of predictions made by machine learning models.
- They provide actionable insights for model evaluation, comparison, and improvement.

Why to use precision and recall instead of accuracy?

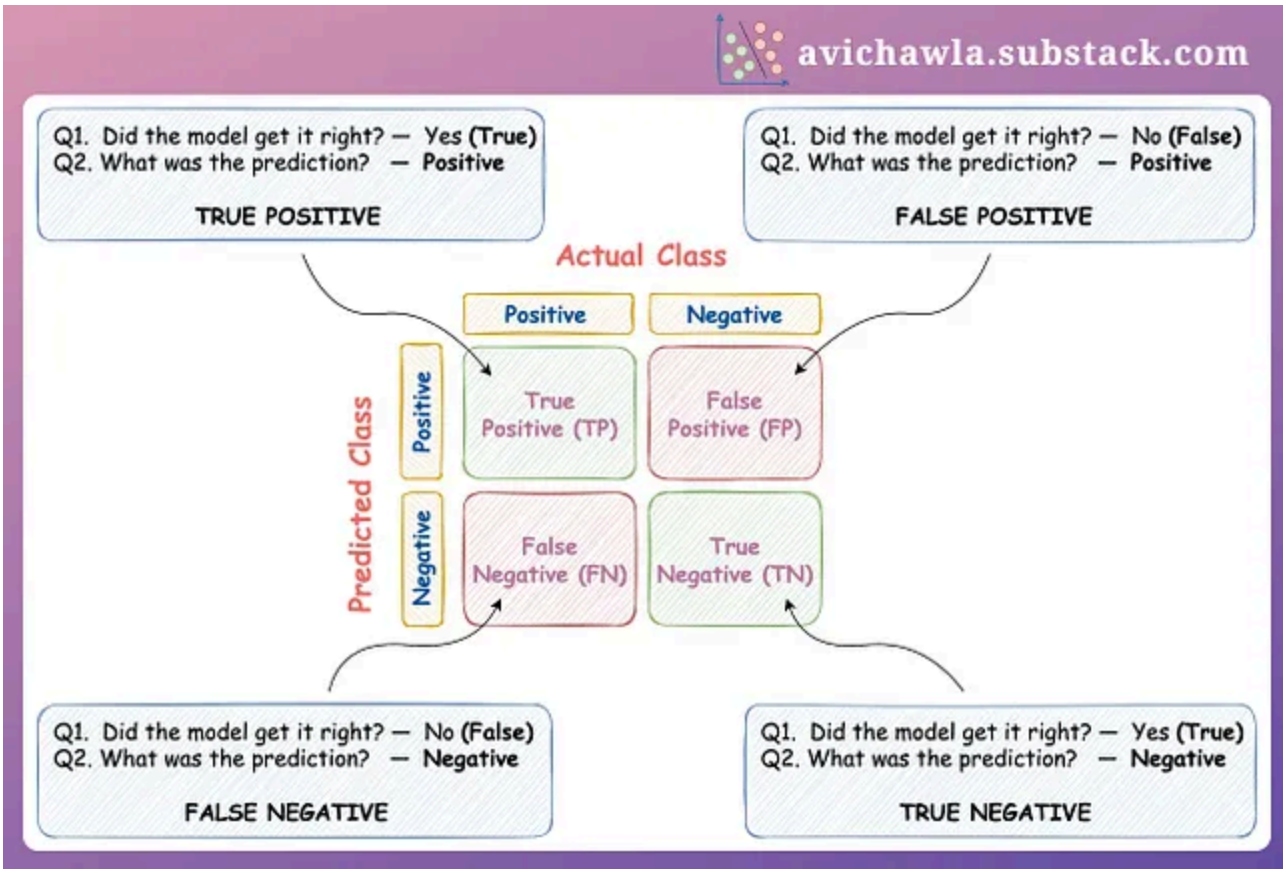
We cannot directly apply accuracy in place of an imbalanced dataset. That's why we need to use precision and recall to properly quanitfy the model's performance.

Let's understand this with an example. if our dataset contains 1000 records out of which 900 belongs to class 1 and 100 belongs to class 2, then if we told our model to predict class 1 everytime,, our models efficiency is going to be 90%.

Applications:

- Performance metrics are used in a wide range of machine learning tasks, including classification, regression, clustering, and recommendation systems.
- They aid in assessing model performance across different domains, such as healthcare, finance, marketing, and natural language processing.

Confusion Matrix



Key Performance Metrics:

1. Precision:

- **Definition:** Precision measures the proportion of true positive predictions among all positive predictions made by the model.
- **Example:** In a spam detection system, precision indicates the percentage of correctly identified spam emails out of all emails classified as spam.
- There are 4 cases in spam classification
- **Case 1:** The mail is a spam and the prediction is, it is a spam mail (TP). Here the models prediction is good
- **Case 2:** The mail is a spam and the prediction is, it is not a spam mail (FN). Here even though, the model's prediction is wrong, it won't cause any issues to the user.
- **Case 3:** The mail is not a spam mail, but the prediction is, it is a spam mail (FP). Here the model's prediction is wrong, and it will cause issues to the user. So here we need to decrease FP, so precision metric is to be used.
- **Case 4:** The mail is not a spam mail and the prediction is, it is not a spam mail (TN)
- Precision tells us out of all the predicted results, how many of them are correctly predicted.

- In precision, our focus is to increase TP and TN and to decrease FP.

$$Precision = \frac{TP}{TP + FP}$$

2. Recall (Sensitivity):

- **Definition:** Recall measures the proportion of true positive predictions among all actual positive instances in the dataset.
- **Example:** In medical diagnosis, recall indicates the percentage of correctly identified patients with a specific disease out of all patients who actually have the disease.
- There are 4 cases in if a person has cancer or no
- **Case 1:** The person has cancer and the prediction is, the person has cancer (TP)
- **Case 2:** The person does not have cancer and the prediction is, that he does not have cancer (TN)
- **Case 3:** The person does not have cancer but the prediction is he does have cancer (FP) which wont cause any issues
- **Case 4:** The person have cancer but the prediction is he does not have cancer (FN) which will cause issues (FN). So here we need to decrease FN so, we need to use recall over here.
- Out of all the actual results, how many of them are correctly predicted results.
- In recall, our focus is to increase TP and to decrease FN

$$Recall = \frac{TP}{TP + FN}$$

NOTE: We need to know the domain expertise when to use precision or recall.

3. Accuracy:

- **Definition:** Accuracy measures the proportion of correct predictions (both true positives and true negatives) among all predictions made by the model.
- **Example:** In sentiment analysis, accuracy indicates the percentage of correctly classified sentiments (positive or negative) out of all predictions made by the model.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

4. F-beta Score:

- **Definition:** The F-beta score is a weighted harmonic mean of precision and recall, where the parameter beta controls the balance between precision and recall.
- **Example:** Tomorrow Stock market is going to crash, here both precision and recall needs to be used as the stock market is important for a company and for an individual.
- Case 1: If FP and FN both are important then, BETA = 1
- Case 2: If FP is more important than FN then, BETA = 0.5
- Case 3: If FN is more important than FP then, BETA = 2

$$F_\beta = (1 + \beta^2) * \frac{(Precision * Recall)}{(\beta^2 * Precision) + Recall}$$

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